



COMPLETE STEP-BY-STEP DEMO

How to Create a Windows Virtual Machine in OCI

Windows Server on Oracle Cloud Infrastructure, from an empty account to a Remote Desktop session, with illustrated screens.

VERSION	1.0
DATE	3 October 2026
TIME NEEDED	About 30-45 minutes
LEVEL	Beginner - no prior OCI experience needed

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1. What you will build

By the end of this demo you will have a **Windows Server 2022** virtual machine running in Oracle Cloud Infrastructure (OCI) that you can use from your own computer over **Remote Desktop (RDP)**, exactly like a physical Windows server in a data centre.

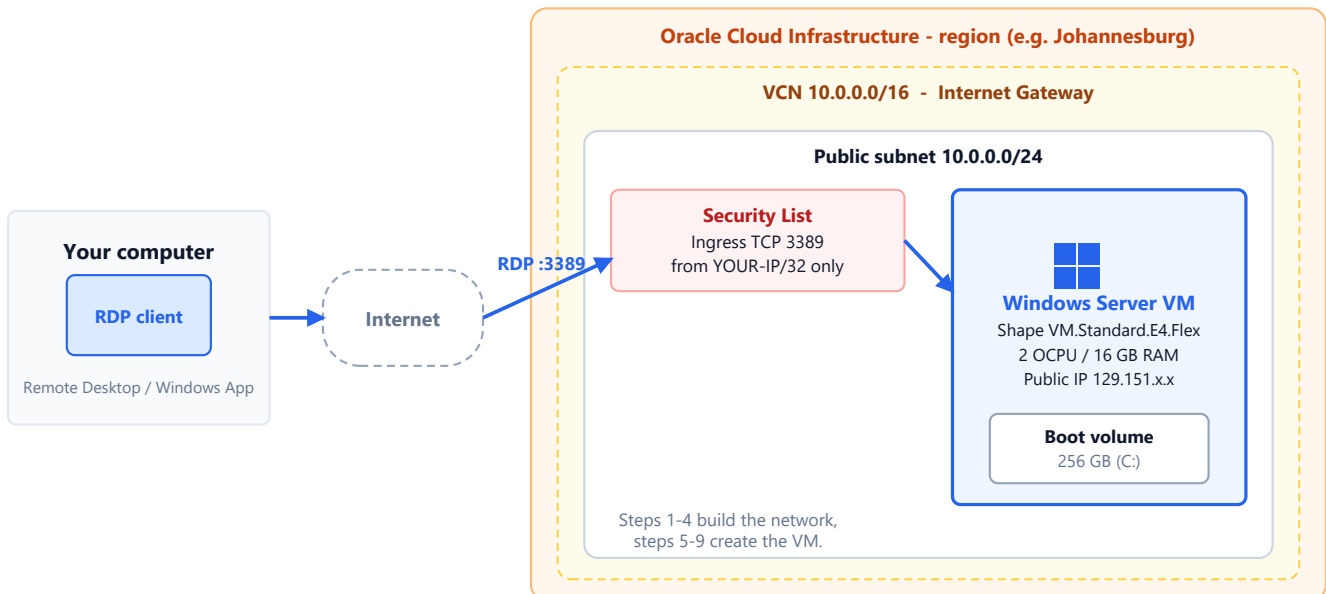


Figure 1. The finished setup: your RDP client reaches the Windows VM over the internet. A Security List rule only lets RDP in from your own IP address. (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Part	What it is	Created in
Compartment	A folder that groups your OCI resources and controls who can manage them.	Step 2 (optional)
VCN + public subnet	Your private network in OCI, with a gateway to the internet.	Step 3
Security List rule	Firewall rule allowing RDP (TCP 3389) from your IP only.	Step 4
Compute instance	The Windows Server virtual machine (CPU, memory, public IP).	Steps 5-9
Boot volume	The VM's C: drive (256 GB by default for Windows images).	Step 8

The 13 steps at a glance

#	Step	#	Step
1	Sign in and choose your region	8	Configure the boot volume and create
2	(Optional) Create a compartment	9	Wait for RUNNING and copy the credentials
3	Create the virtual cloud network (VCN)	10	Connect with Remote Desktop
4	Allow RDP from your IP address	11	Change the initial password
5	Start creating the instance	12	Secure and update Windows
6	Choose the Windows Server image	13	Stop or delete the VM to control cost
7	Choose a shape and networking		

2. Before you begin

You need	Details
An OCI account that can be billed	Windows Server VMs are not part of the Always Free tier : Microsoft licensing is charged per OCPU per hour while the VM runs. A new Free Trial account's trial credits can pay for it, otherwise upgrade to Pay As You Go .
Permission to create resources	If you are not the tenancy administrator, ask for a policy such as <code>Allow group <your-group> to manage all-resources in compartment <name></code> .
Your public IP address	Search "what is my IP" in your browser and note it, e.g. <code>203.0.113.25</code> . You will allow RDP only from this address.
A Remote Desktop client	Windows: built-in Remote Desktop Connection (<code>mstsc</code>). macOS / iOS / Android: Microsoft Windows App (formerly Microsoft Remote Desktop). Linux: Remmina.

Cost. Billing for OCPU, memory and the Windows licence runs while the instance is **RUNNING**. Storage for the boot volume is billed until you delete it. Use OCI's Cost Estimator (oracle.com/cloud/costestimator.html) for current prices, and see Step 13 to stop or delete the VM when you're done with the demo.

Windows needs an x86 shape. Windows images run only on AMD or Intel shapes, such as VM.Standard.E4.Flex, E5.Flex or Standard3.Flex. They can't use the Ampere (Arm) A1 shapes that are popular for free Linux VMs.

3. Step-by-step demo

Step 1 - Sign in and choose your region

1. Go to **cloud.oracle.com**, enter your **Cloud Account Name** (tenancy), click **Next** and sign in.
2. Look at the **region** name at the top right of the console. Choose the region closest to the people who will use the VM, e.g. **South Africa Central (Johannesburg)**. Everything you create lives in this region.
3. Open the navigation menu (the three lines at the top left). You will use **Networking** in Step 3 and **Compute > Instances** in Step 5.

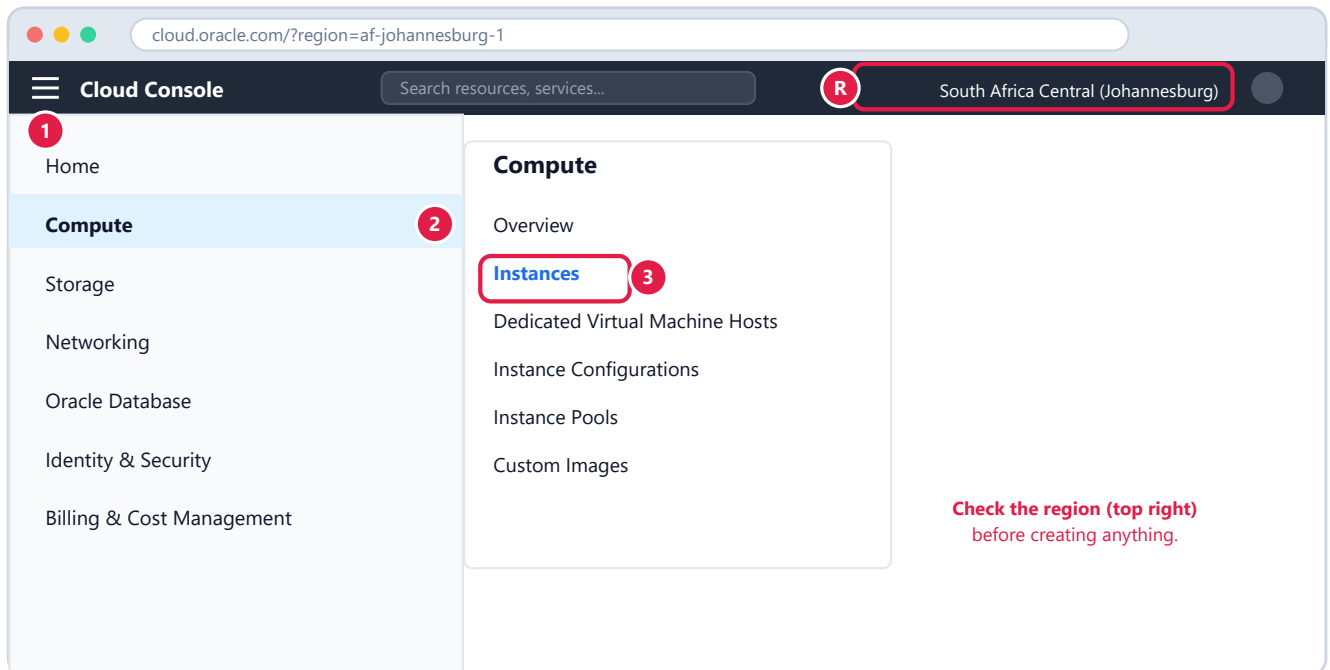


Figure 2. Navigation menu: (1) open the menu, (2) Compute, (3) Instances. Confirm the region at the top right first. (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Step 2 - (Optional) Create a compartment

Compartments keep projects separate and make clean-up easy (delete everything inside one folder). Skip this step to use the default root compartment.

1. Menu > **Identity & Security** > **Compartments** > **Create Compartment**.
2. Name: `fleet-demo`; Description: `Windows VM demo`; Parent: your root compartment. Click **Create Compartment**.
3. In every later screen, pick `fleet-demo` in the **Compartment** drop-down (usually on the left side of list pages).

Step 3 - Create the virtual cloud network (VCN)

The VM needs a network with a route to the internet. The VCN Wizard builds all of it in one go.

1. Menu > **Networking** > **Virtual Cloud Networks**. Make sure your compartment is selected.
2. Click **Start VCN Wizard**, choose **Create VCN with Internet Connectivity** and click **Start VCN Wizard**.

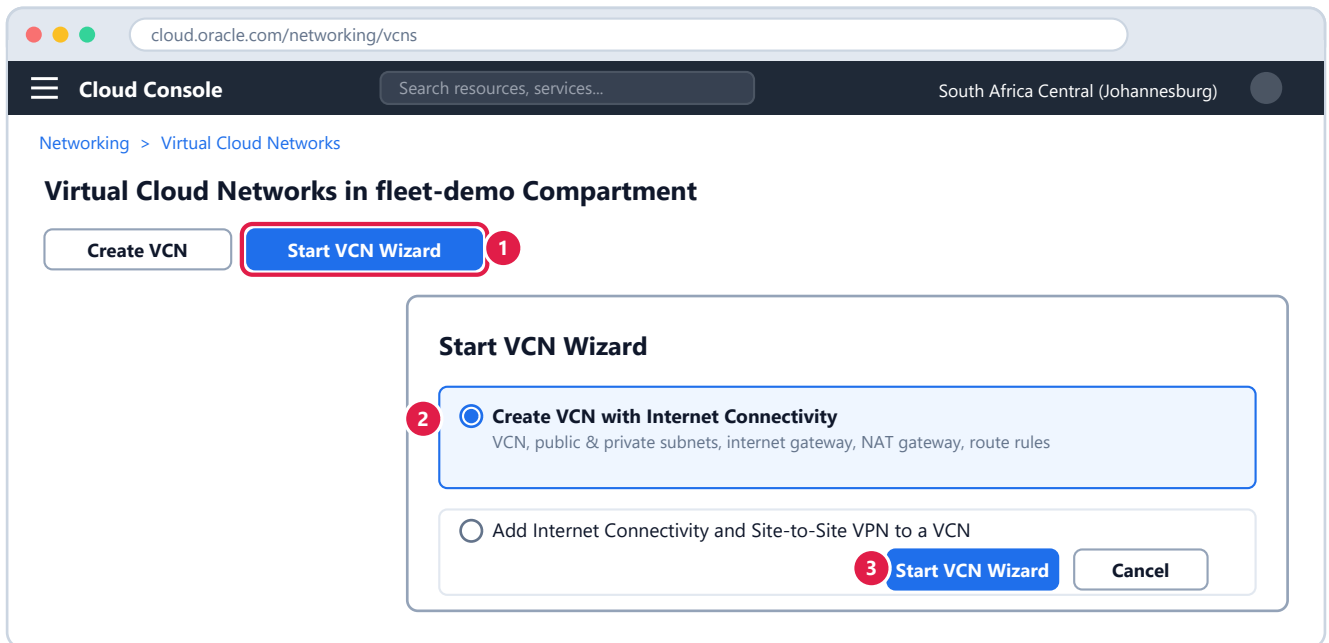


Figure 3. Start the VCN Wizard (1), choose 'Create VCN with Internet Connectivity' (2), confirm (3). (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

3. Enter **VCN name** `windows-demo-vcn`. Leave the CIDR blocks at their defaults (VCN `10.0.0.0/16`, public subnet `10.0.0.0/24`, private subnet `10.0.1.0/24`).
4. Click **Next**, review, then click **Create**. Wait until every item shows a green tick, then click **View VCN**.

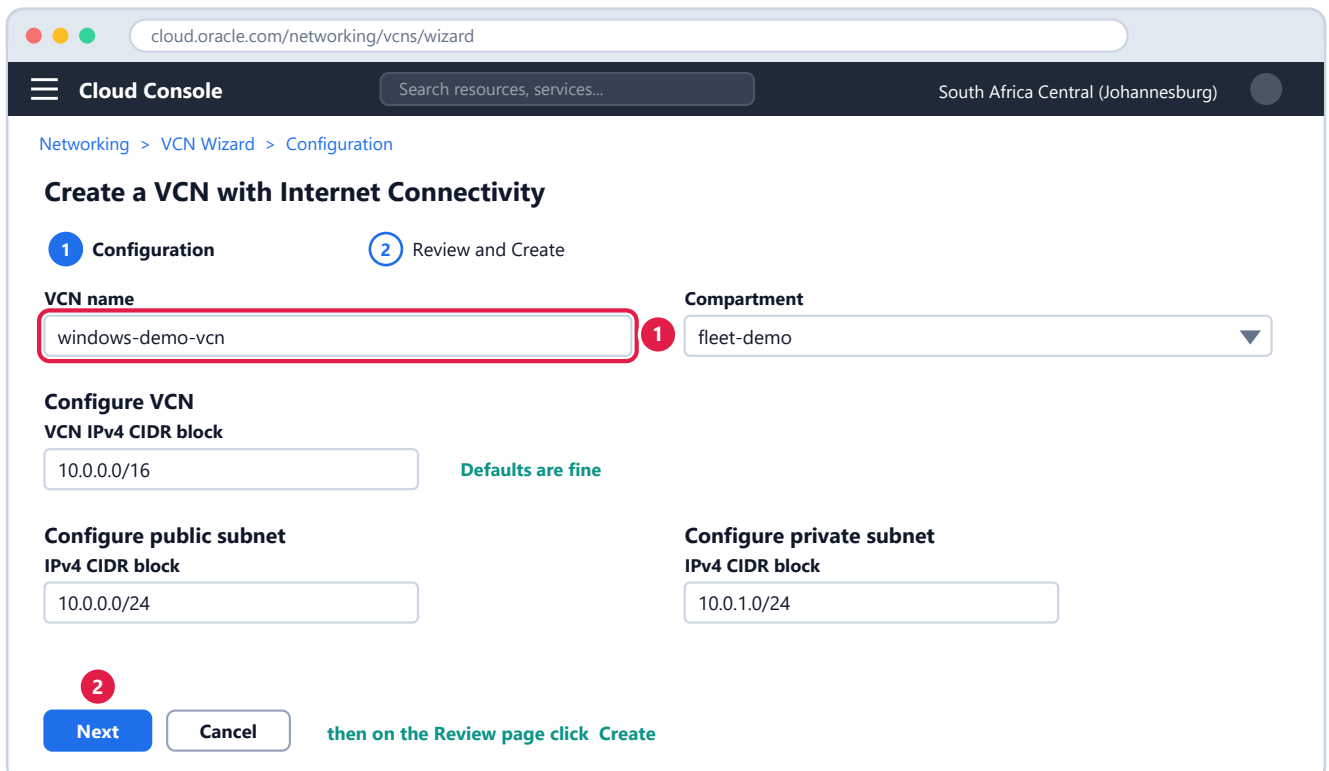


Figure 4. Name the VCN (1), keep the default address ranges, click Next (2) and then Create. (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Step 4 - Allow RDP from your IP address

By default the subnet only allows SSH (port 22). Windows Remote Desktop uses **TCP port 3389**, which you must open, but only for your own address.

1. On the VCN page open **Security Lists** (in the left menu or the **Security** tab) and click **Default Security List for windows-demo-vcn**.
2. Under **Ingress Rules** click **Add Ingress Rules** and fill in the values below, then click **Add Ingress Rules**.

Field	Value
Stateless	Unticked
Source Type	CIDR
Source CIDR	<your-public-ip>/32, e.g. 203.0.113.25/32 (the /32 means "exactly this one address")
IP Protocol	TCP
Source Port Range	Leave empty (All)
Destination Port Range	3389
Description	Allow RDP from my office only

cloud.oracle.com/networking/vcns/.../security-lists

Cloud Console Search resources, services... South Africa Central (Johannesburg)

Networking > VCNs > windows-demo-vcn > Security Lists > Default Security List

Add Ingress Rules

Ingress Rule 1

☐ Stateless

Source Type
CIDR

Source CIDR
203.0.113.25/32

IP Protocol
TCP

Source Port Range (optional)
All

Destination Port Range
3389

Description
Allow RDP from my office only

Add Ingress Rules **Cancel**

Never use 0.0.0.0/0
for port 3389. Open RDP is scanned and brute-forced within minutes.

Figure 5. Ingress rule for RDP: your IP /32 (1), TCP (2), port 3389 (3), then Add Ingress Rules (4). (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Never open 3389 to 0.0.0.0/0. Bots scan the whole internet for open RDP ports and start guessing passwords within minutes. If your IP changes (home internet, travel), edit this rule rather than opening it to everyone. You can also delete the default SSH (22) rule; Windows doesn't need it.

Step 5 - Start creating the instance

1. Menu > **Compute** > **Instances**, check the compartment, click **Create instance**.
2. **Name**: win-demo-01. **Create in compartment**: fleet-demo.
3. **Placement**: keep the suggested **Availability domain** (any works).
4. In **Image and shape**, click **Change image**.

The illustration shows the OCI Console interface for creating a compute instance. The browser address bar shows 'cloud.oracle.com/compute/instances/create'. The console header includes the 'Cloud Console' logo, a search bar, and the region 'South Africa Central (Johannesburg)'. The breadcrumb trail is 'Compute > Instances > Create compute instance'. The left sidebar lists five steps: 1. Basic information, 2. Security, 3. Networking, 4. Storage, and 5. Review. The main content area is titled 'Basic information' and contains three sections: 'Name' with a text input field containing 'win-demo-01' (highlighted with a red box and a red circle with the number 1), 'Create in compartment' with a dropdown menu showing 'fleet-demo', and 'Placement' with a dropdown menu showing 'AD-1'. Below these is the 'Image and shape' section, which shows 'Image: Oracle Linux 9 (default)' and 'Shape: VM.Standard.E4.Flex'. A red box and a red circle with the number 2 highlight the 'Change image' button in the 'Image and shape' section.

Figure 6. Basic information: name the instance (1), then Change image (2). (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Step 6 - Choose the Windows Server image

1. On the **Platform images** tab select **Windows**.
2. In **Image build** choose **Windows Server 2022 Standard** (or the newest Windows Server version listed in your region).
3. Read and accept the **Microsoft license terms** checkbox.
4. Click **Select image**.

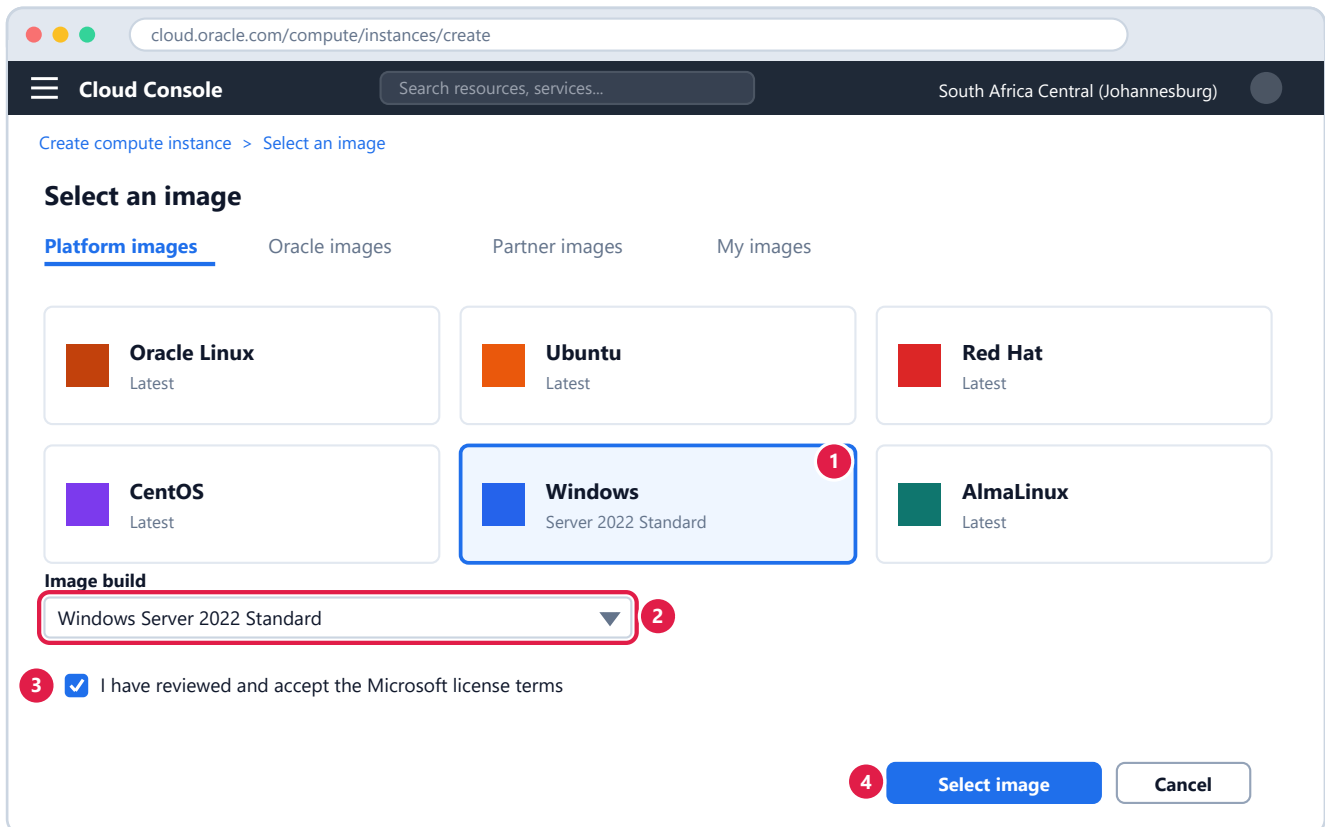


Figure 7. Select Windows (1), pick the Windows Server build (2), accept the licence (3), Select image (4). (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Licence included. Oracle's Windows platform images include the Windows Server licence; you don't need your own product key. The licence cost is part of the hourly price while the VM runs.

Step 7 - Choose a shape and networking

The **shape** is the size of the VM. Click **Change shape**, then:

1. Instance type **Virtual machine**, shape series **AMD** (Ampere is not available for Windows).
2. Select **VM.Standard.E4.Flex** (or E5.Flex).
3. Set **Number of OCPUs** to **2** and **Amount of memory** to **16 GB**, then click **Select shape**.

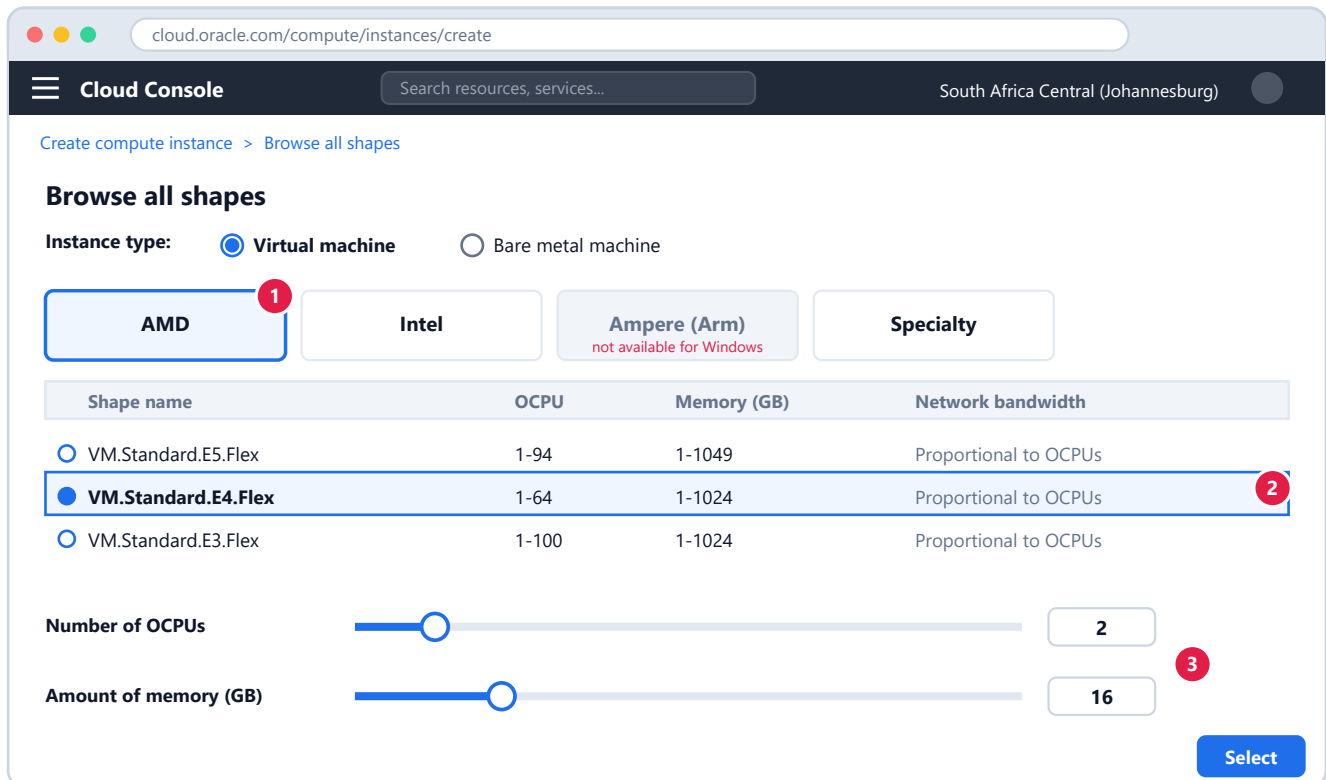


Figure 8. AMD series (1), VM.Standard.E4.Flex (2), 2 OCPUs and 16 GB memory (3). (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Use	OCPUs	Memory	Notes
Quick demo / light admin tasks	1	8-16 GB	Cheapest; Windows feels slow below 8 GB.
Recommended for this demo	2	16 GB	Comfortable RDP desktop, small apps.
Small application server	4	32 GB	IIS, SQL Server Express, build agents.

One AMD **OCPU** equals two vCPUs (threads), so 2 OCPUs appear as 4 processors inside Windows.

Now scroll to **Networking** (Primary VNIC information):

1. Choose **Select existing virtual cloud network** and pick `windows-demo-vcn`.
2. Choose **Select existing subnet** and pick the **public subnet** (not the private one).
3. Make sure **Automatically assign public IPv4 address** is ticked.

cloud.oracle.com/compute/instances/create

Cloud Console Search resources, services... South Africa Central (Johannesburg)

Create compute instance > Networking

Primary VNIC information

1 ☒ Select existing virtual cloud network ☐ Create new virtual cloud network

Virtual cloud network in fleet-demo: windows-demo-vcn

Subnet in fleet-demo: public subnet-windows-demo-vcn (regional) 2

Primary VNIC IP addresses

☒ Automatically assign private IPv4 address

☒ Automatically assign public IPv4 address 3

Add SSH keys
Not used for Windows: you sign in with the 'opc' user and the initial password shown on the instance page after it is created (Step 9).

Figure 9. Existing VCN (1), the public subnet (2), and a public IPv4 address (3). SSH keys are not used for Windows. (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Step 8 - Configure the boot volume and create

1. In **Boot volume**, keep the default size (**256 GB** for Windows images) and leave **in-transit encryption** on.
2. Check the **Review** summary: Windows image, E4.Flex shape, public subnet, public IP.
3. Click **Create**. The instance appears with the state **PROVISIONING** (orange).

cloud.oracle.com/compute/instances/create

Cloud Console Search resources, services... South Africa Central (Johannesburg)

Create compute instance > Storage / Review

Boot volume

☐ Specify a custom boot volume size and performance setting

Default boot volume size for this Windows image: 256 GB (Balanced, 10 VPU/GB)

☒ Use in-transit encryption

☐ Encrypt this volume with a key that you manage

Review

Name	win-demo-01
Image	Windows Server 2022 Standard
Shape	VM.Standard.E4.Flex - 2 OCPU, 16 GB
Subnet	public subnet-windows-demo-vcn
Public IPv4	Yes

Create 1 **Save as stack** **Cancel**

Estimated cost is shown at the bottom of the page

Figure 10. Keep the default boot volume, review the summary and click Create (1). (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Step 9 - Wait for RUNNING and copy the credentials

1. Wait until the state changes to **RUNNING** (green), usually 2-5 minutes.
2. On the **Details** tab find **Instance access**. Copy the **Public IP address**.
3. Note the **Username** (`opc`) and click **Show / Copy** next to **Initial password**.

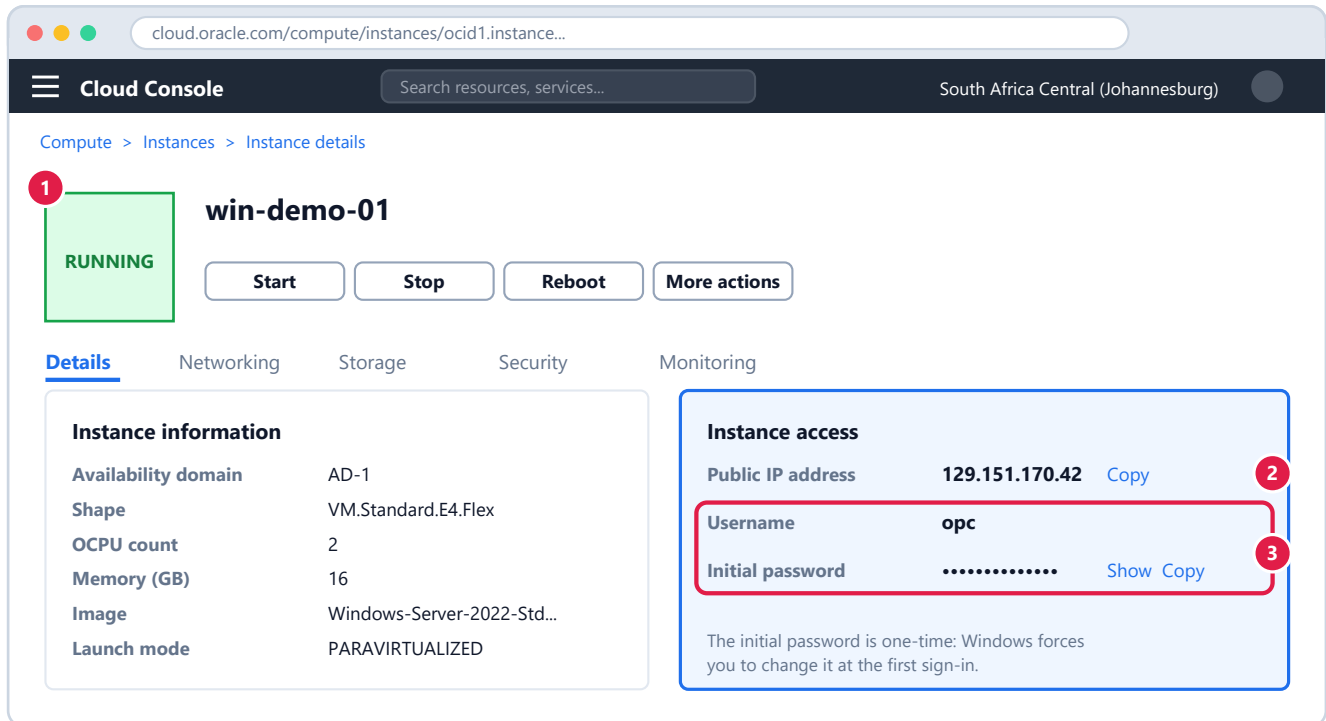


Figure 11. RUNNING (1), the public IP (2), and the one-time username/password (3) under Instance access. (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

Give Windows a few more minutes. RUNNING means the VM has booted, but Windows still finishes its first-boot setup (sysprep, network, RDP service) for another 3-10 minutes. If RDP doesn't connect straight away, wait and try again.

Step 10 - Connect with Remote Desktop

On Windows: press **Win + R**, type `mstsc` and press Enter (or search for Remote Desktop Connection).

1. In **Computer** paste the public IP and click **Connect**.
2. In the credentials window choose **More choices > Use a different account**, enter user `opc` and the initial password, click **OK**.
3. A certificate warning appears because the VM uses a self-signed certificate. Check the name matches your instance and click **Yes**.

Remote Desktop Connection

Remote Desktop
Connection

Computer:
129.151.170.42

User name: None specified
You will be asked for credentials when you connect.

2 **Connect** **Help**

Windows Security

Enter your credentials
These credentials will be used to connect to 129.151.170.42.

opc

.....

[More choices](#) > [Use a different account](#)

4 **OK** **Cancel**

Figure 12. Remote Desktop Connection: enter the public IP (1), Connect (2), sign in as opc with the initial password (3), OK (4).
(Illustration of the OCI Console; labels match the console, layout may differ slightly.)

```
:: Windows shortcut - opens Remote Desktop straight to the VM
mstsc /v:129.151.170.42
```

Your computer	RDP client	How
Windows 10/11	Remote Desktop Connection	<code>mstsc /v:<public-ip></code>
macOS / iPhone / iPad	Windows App (App Store)	Add PC > PC name = public IP > user <code>opc</code>
Android	Windows App (Play Store)	Add PC > public IP
Linux	Remmina	New connection, protocol RDP, server = public IP

Step 11 - Change the initial password

1. Accept the certificate prompt (**Yes**).
2. Windows shows **"The user's password must be changed before signing in"**. Click **OK**.
3. Enter the initial password as the old password, then a new strong password twice. Windows requires at least 12 characters with upper case, lower case, numbers and symbols. Store it in a password manager.

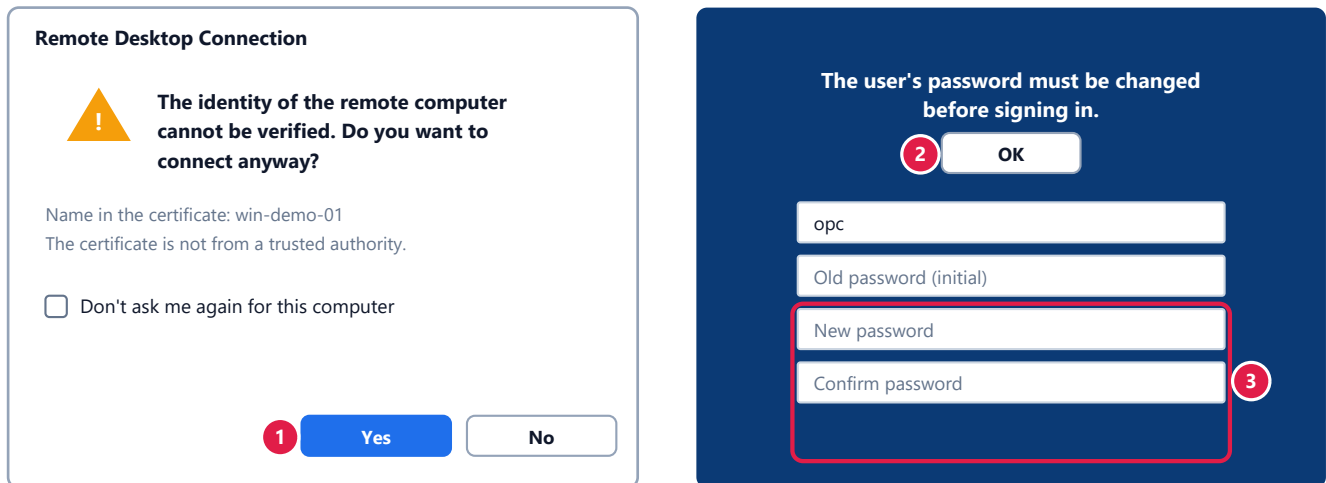


Figure 13. Accept the self-signed certificate (1), acknowledge the password change (2), set a new password (3). (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

After the password change, the Windows Server desktop and **Server Manager** open. Your Windows VM is ready.

Step 12 - Secure and update Windows

- **Windows Update:** Settings > Windows Update > **Check for updates**, install everything and restart (your RDP session drops; reconnect after 2-3 minutes).
- **Separate admin account:** create your own administrator user (Computer Management > Local Users and Groups) instead of using `opc` every day.
- **Keep RDP restricted:** if your IP changes, update the Security List rule from Step 4; never open it to everyone.
- **Keep the IP fixed (optional):** the public IP is ephemeral and changes if the instance is recreated. To keep it, go to the instance's **Attached VNICS > IPv4 Addresses** and convert it to a **Reserved public IP**.
- **Backups:** open the instance's **Boot volume** and assign a **backup policy** (e.g. Bronze/Silver/Gold) so OCI takes scheduled backups of C:.

```
# In PowerShell on the VM (Run as Administrator) - useful first commands
Get-ComputerInfo -Property OsName, OsVersion, CsNumberOfLogicalProcessors, CsTotalPhysicalMemory
Get-NetFirewallRule -DisplayGroup "Remote Desktop" | Select DisplayName, Enabled
Set-TimeZone -Id "South Africa Standard Time"
```

Step 13 - Stop or delete the VM to control cost

Action	What happens	Billing
Stop	Windows shuts down; VM keeps its disk and configuration.	OCPU, memory and Windows licence charges stop. Boot volume storage continues.
Start	VM boots again (the ephemeral public IP may change unless reserved).	Charges resume.
Terminate	VM is deleted. Tick Permanently delete the attached boot volume to delete C: too.	All charges stop (once the boot volume is deleted).

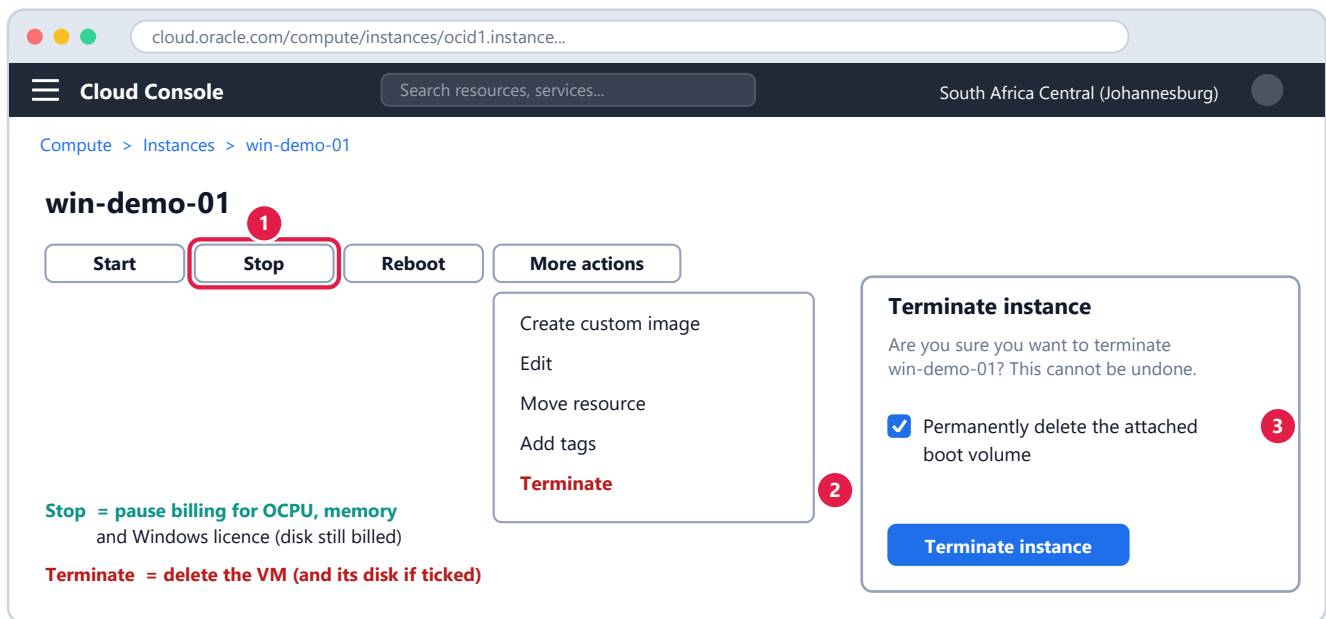


Figure 14. Stop (1) pauses compute and licence billing; More actions > Terminate (2) deletes the VM, optionally with its boot volume (3). (Illustration of the OCI Console; labels match the console, layout may differ slightly.)

To remove the whole demo: terminate the instance (with its boot volume), then delete `windows-demo-vcn` (Networking > VCN > **Delete**, which also removes its subnets, gateways and security list), and finally the `fleet-demo` compartment if you created one.

4. Troubleshooting

Problem	Likely cause	Fix
RDP: "Remote Desktop can't connect to the remote computer"	Port 3389 not allowed from your current IP; Windows still finishing first boot	Check the Step 4 rule matches your current public IP; wait 5-10 minutes after RUNNING and retry.
Worked yesterday, fails today	Your home/office public IP changed	Update the Source CIDR in the ingress rule.
"Out of host capacity" when creating	No free capacity for that shape in that AD	Pick another availability domain or another AMD/Intel shape (E5.Flex, Standard3.Flex).
Windows not offered for the shape	An Ampere (Arm) shape is selected	Choose an AMD or Intel shape (Step 7).
Initial password not shown	Instance still provisioning, or it has been changed already	Wait for RUNNING. If lost after changing it, use the instance Console connection to reset it, or recreate.
"Service limit exceeded"	Tenancy quota for that shape is 0 or used up	Governance & Administration > Limits, Quotas and Usage > request a limit increase.
Very slow desktop	Too little memory/OCPU, or updates installing	Resize: instance > Edit > shape (stop it first for some changes).
Can't find the instance	Wrong region or compartment selected	Check the region (top right) and the compartment filter.

5. Quick reference card

Item	Value used in this demo
Region	South Africa Central (Johannesburg)
Compartment	<code>fleet-demo</code>
VCN / subnet	<code>windows-demo-vcn</code> 10.0.0.0/16, public subnet 10.0.0.0/24
Ingress rule	TCP 3389 from <code><your-ip>/32</code>
Instance name	<code>win-demo-01</code>
Image	Windows Server 2022 Standard (platform image, licence included)
Shape	VM.Standard.E4.Flex, 2 OCPU / 16 GB
Boot volume	256 GB (default)
Sign-in	User <code>opc</code> + initial password (Instance access), changed at first login
Connect	<code>mstsc /v:<public-ip></code>

Checklist

Done	Task
☐	Correct region and compartment selected
☐	VCN created with the wizard (internet connectivity)
☐	Ingress rule TCP 3389 from my IP /32 only
☐	Instance created: Windows Server image, AMD/Intel shape, public subnet, public IPv4
☐	Instance RUNNING; public IP and initial password copied
☐	Connected with RDP and changed the initial password
☐	Windows Update run; boot volume backup policy assigned
☐	VM stopped or terminated when not needed